

SHEDDING



Lecture learning outcomes

At the end of this lecture the student should:

- 1- State what is the meaning of shedding
- 2- Know the mechanism of shedding
- 3- Explain the accompanying histological events
- 3- Recognize the clinical considerations of shedding (shedding abnormalities)

Shedding of Primary Teeth

Definition:

- Physiologic elimination(loss) of deciduous teeth due to progressive root resorption due to the influence of the pressure exerted by the erupting permanent successor



Initiation of shedding

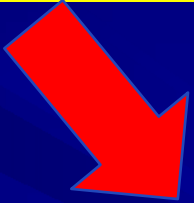
- Two factors initiate shedding of the primary teeth :
 - a. **Local factors:**
 - Pressure factor** which leads to the differentiation of the odontoclasts
 - * pressure of the erupting permanent tooth
 - * increase of the masticatory forces applied to teeth
 - b. **Genetic factor**
 - is probably responsible for the initiation of root resorption at specific time.(evidenced by the programmed apoptosis in periodontal ligament)

Pattern of Shedding

According to Developmental position of permanent successor
pattern of root resorption takes place



1-Primary
Anterior
teeth



2- Primary
molars

1-Primary Anterior teeth

- The permanent anteriors develop in **apical & lingual** position to their deciduous predecessors.

-Then they move into an

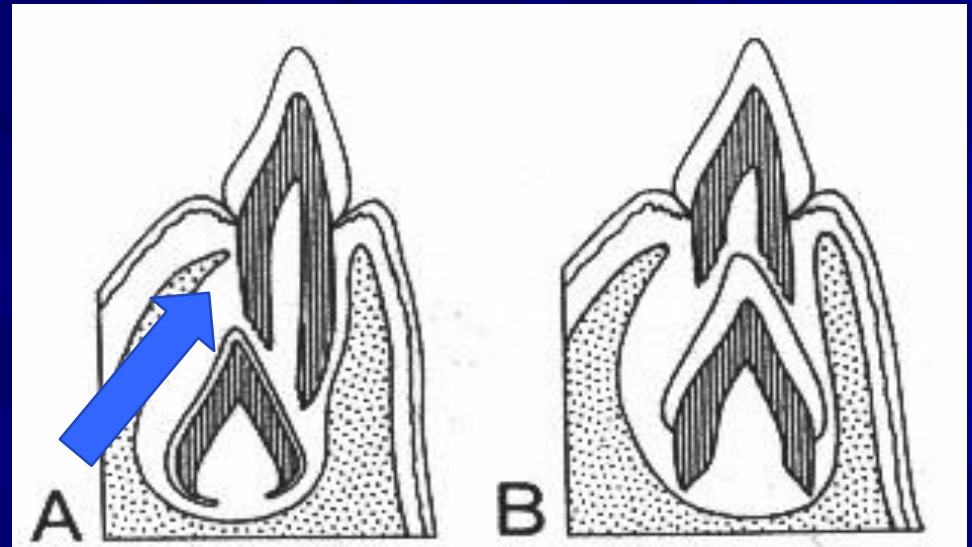
incisolabial direction,

-In latter stages,

they are frequently located **apical** to their deciduous predecessors .

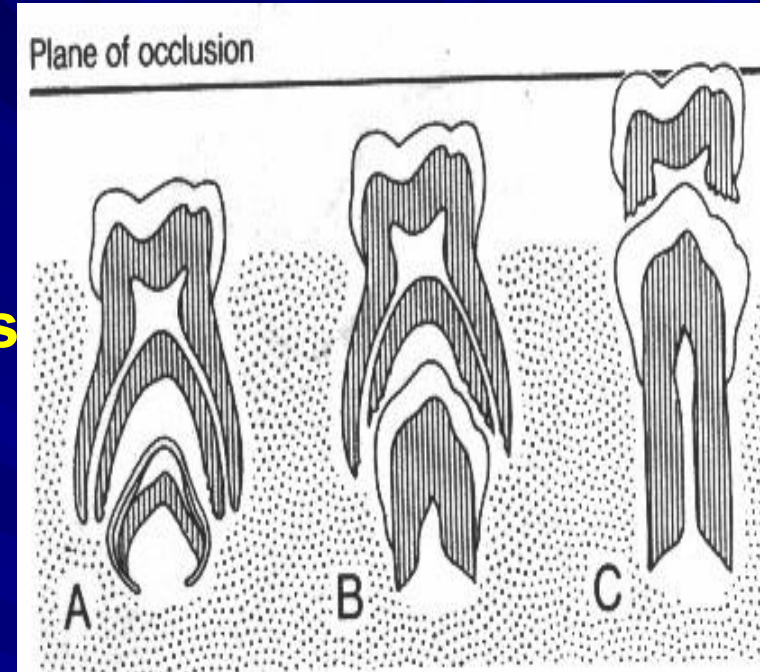
- Thus, the resorptive process begins on **apico-lingual** root surface, and then proceeds in a horizontal plane apically

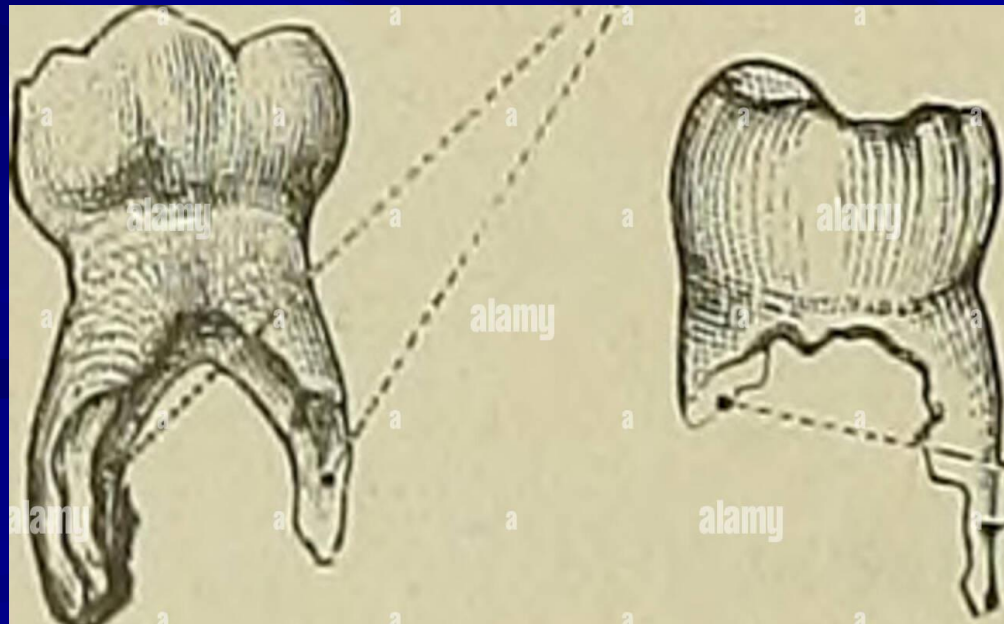
-This help eruption of perm.tooth in exact position of their predecessors



2- Primary molars

- The premolars begin their development **lingual** to their corresponding primary molars .
- In later stage, they are frequently **found between the divergent roots** of the primary molars .
- Therefore resorption of the roots of deciduous molars first **begins on their inner surfaces (inter-radicular)**









Histology of shedding

The resorption of deciduous teeth is achieved by:

Odontoclasts.....involved in removal of hard dental tissues (dentin, cementum and even enamel)

Origin: have the same origin of osteoclasts , that is

1- monocytes .

or 2- undifferentiated mesenchymal cells .

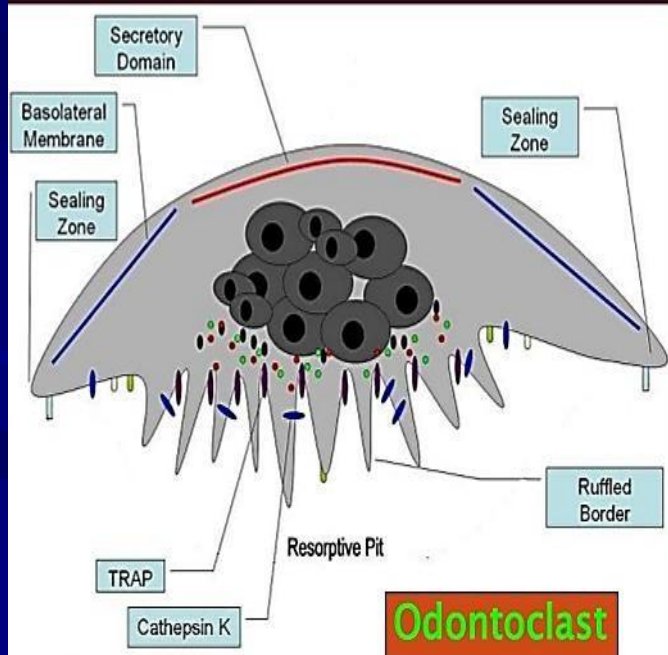
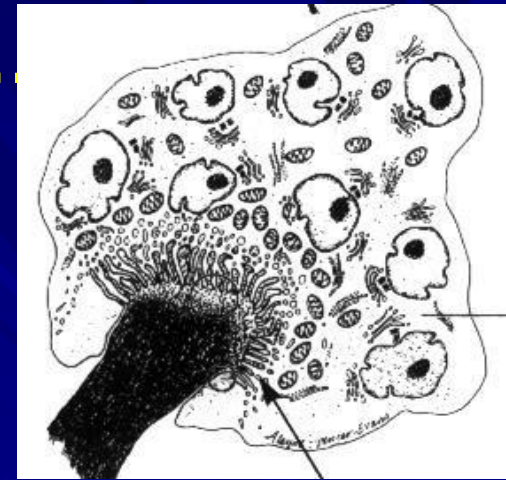
Site.....Odontoclasts are found on

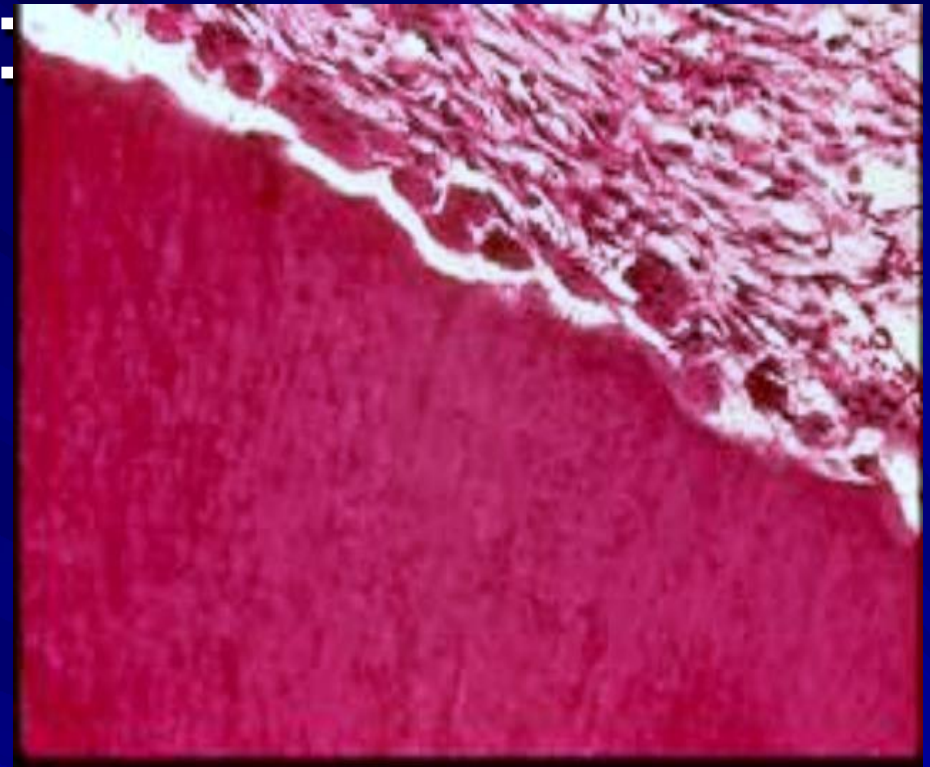
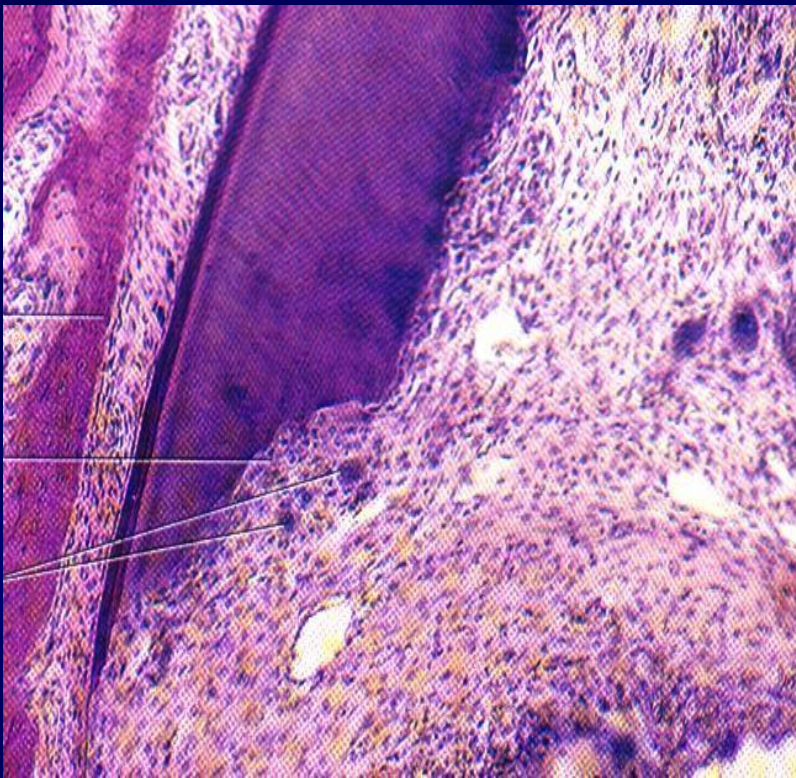
1- surface of roots being resorbed,

2- may be found in root canal and pulp chambers of resorbing deciduous molars.

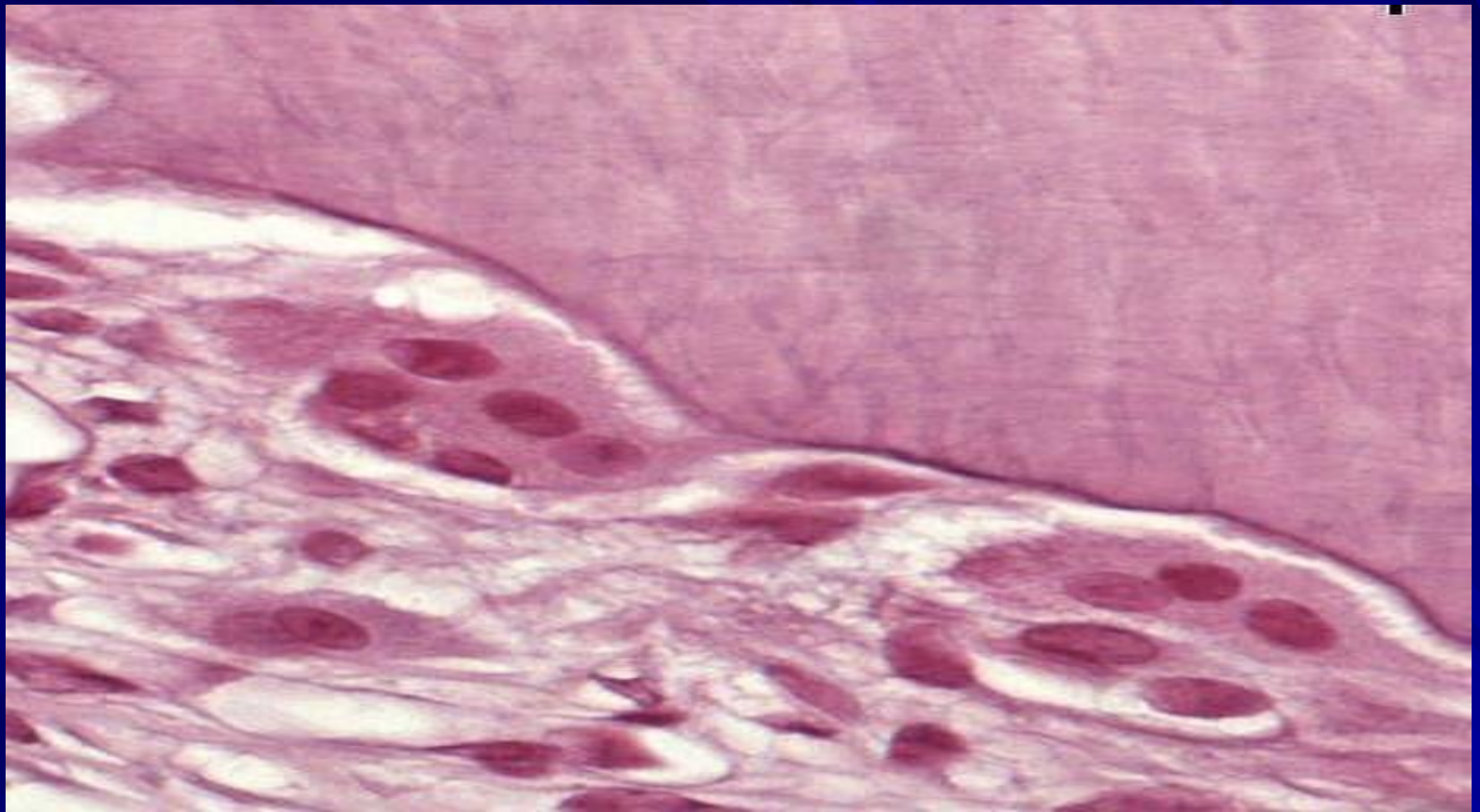
Shape and structure of odontoclasts.
are identical to osteoclasts;
It is a large cell that is characterized by:

- Multiple nuclei
- Cytoplasm with "foamy" appearance.
- "Ruffled border" (brush border) which faces the surface being resorbed





- Odontoclasts are stimulated by the pressure from erupting permanent teeth.
- These cells line up the root surface and appear occupying hollowed-out shallow depressions known as **Howship's lacunae** (scalloped shaped areas of root).



■ **Process of shedding (resorption)**

1- Mineral removal: during resorption, the minerals of the dental hard tissues are primarily removed by acids secreted by odontoclasts

2- The disintegration of the organic matrix occurs by secretion of the proteolytic enzymes by odontoclast

i.e.: 1stdemineralization (by acids)

Then 2nd...organic matrix degradation(by proteolytic enzymes)

Histological features of teeth undergoing shedding in

(Root – Periodontal lig.-Pulp)

1- Root surfaces

- exhibit resorption lacunae
- odontoclast cells are often seen within these concavities

2 Periodontal fibroblasts in the area

- Show signs of impaired function with abrupt loss of P.L

without signs of inflammation.

- Programmed cell death (apoptosis) is seen during shedding.



3- The pulp tissue

- the pulp appears histologically normal except that neural elements seem to be missing.
- Thus the pulp does not contribute to the process of shedding and plays a passive role in this process.

To Summarize

Mechanism of resorption and shedding:

A) Resorption of hard dental tissues.

both osteoclasts and odontoclasts
....resorb bone and hard dental tissues.

B) Resorption of dental soft tissues.

* PDL.....Programmed cell death
(Apoptosis)

*Pulp.....play passive role.

Clinical considerations

Abnormalities of shedding

1. Submerged deciduous teeth

- Stoppage of active eruption of the affected teeth and they become at a lower level than adjacent teeth

Can be caused by ankylosis

Ankylosis is:

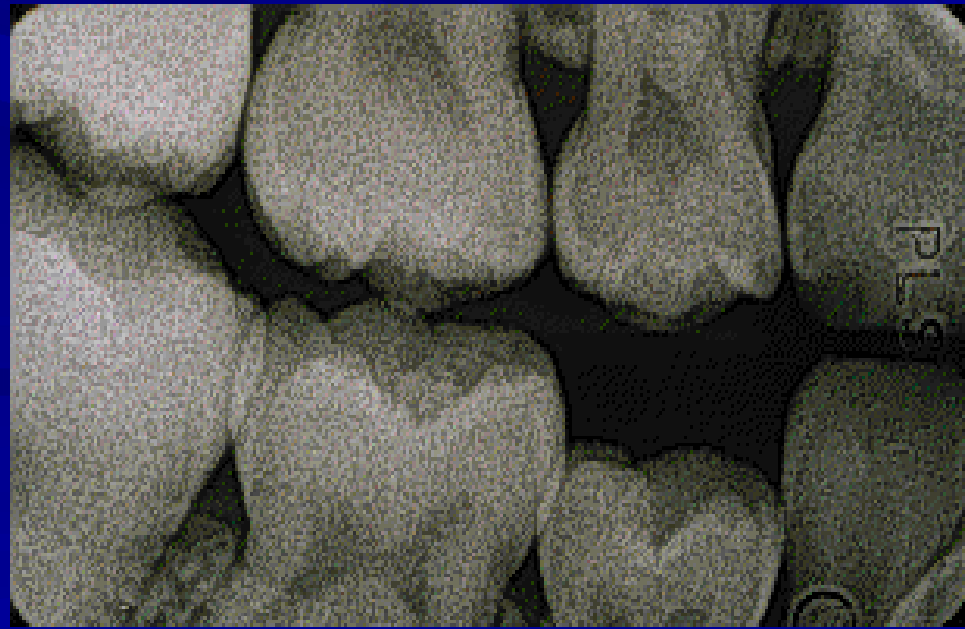
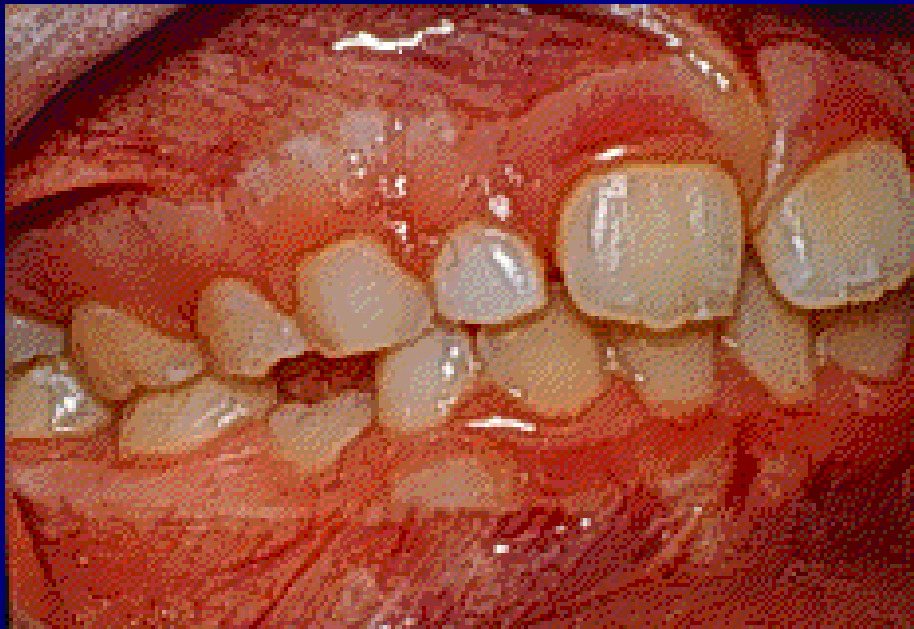
a bony union between the alveolar bone and the roots of teeth.



Causes :

- 1- mild trauma cause damage of PDL
That leads to ankylosis of deciduous instead of loss.
- 2- intermittent resorption of roots

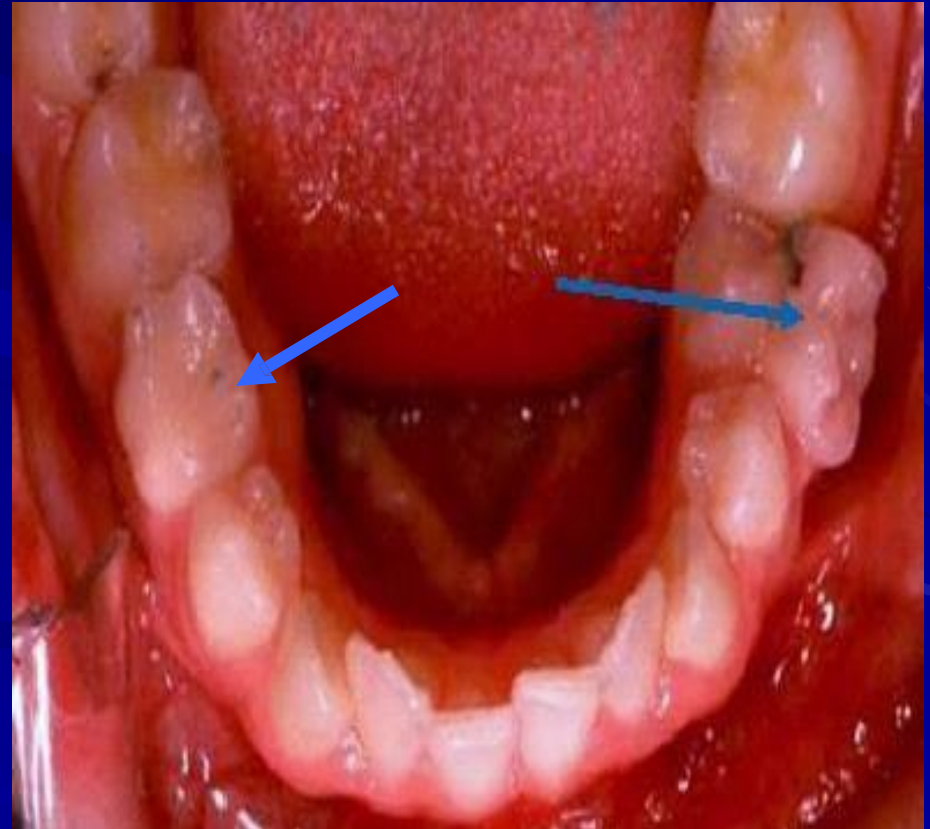
N.B.: the submerged teeth showed be removed



2. Retained deciduous teeth

-Persistence of the primary teeth beyond the age at which they are normally shed and replaced by the permanent teeth

Most common in upper B and lower E



It appears normal and stay functioning for a while then it will be lost due to increased masticatory forces

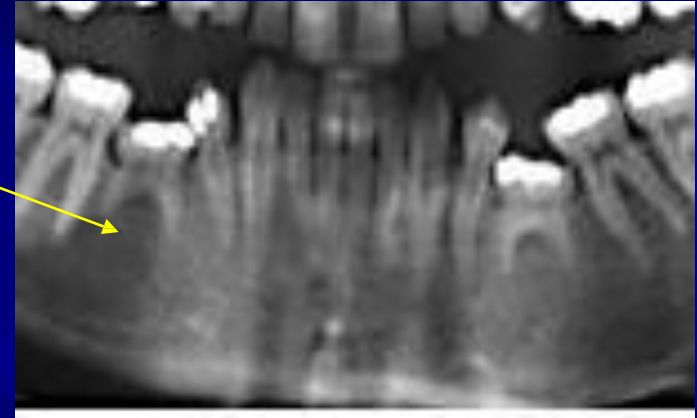


Retained deciduous lateral incisor with
permanent canine

The Causes of retained deciduous teeth

1- Congenitally Missing Teeth (Hypodontia) –

- Usually occurs with lateral incisors & second premolars



Missing teeth!!!

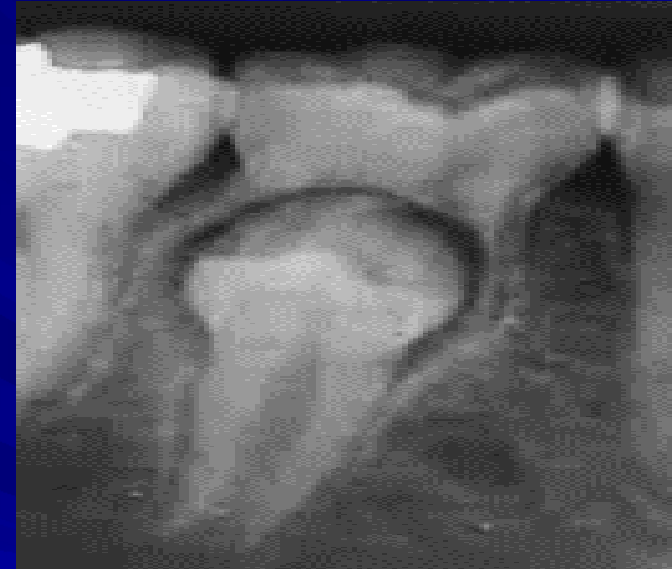
2- Impaction of permanent teeth



3- Ankylosis of permanent tooth due trauma

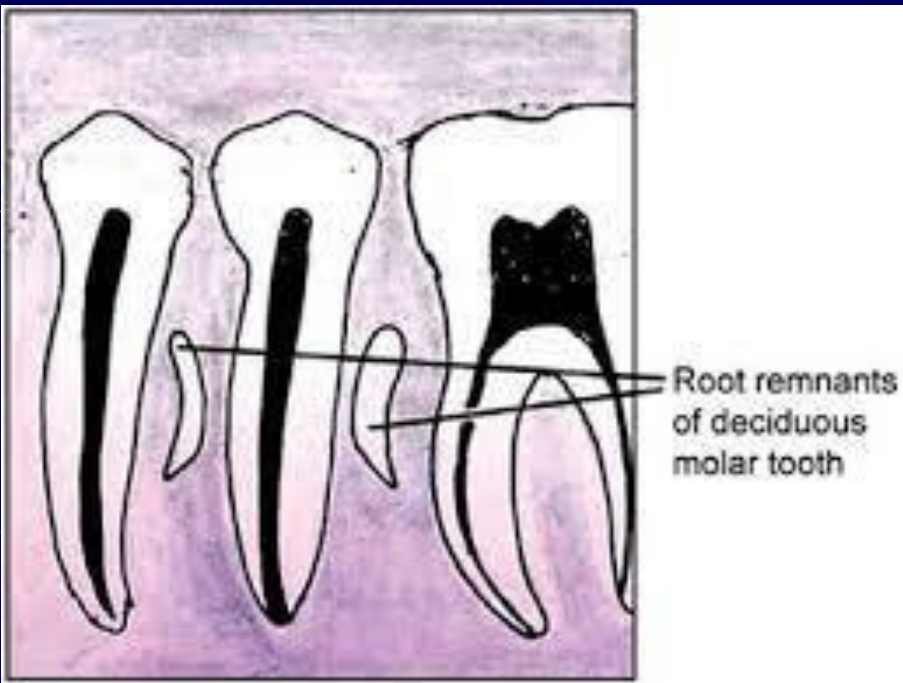
3- Remnants of deciduous teeth:

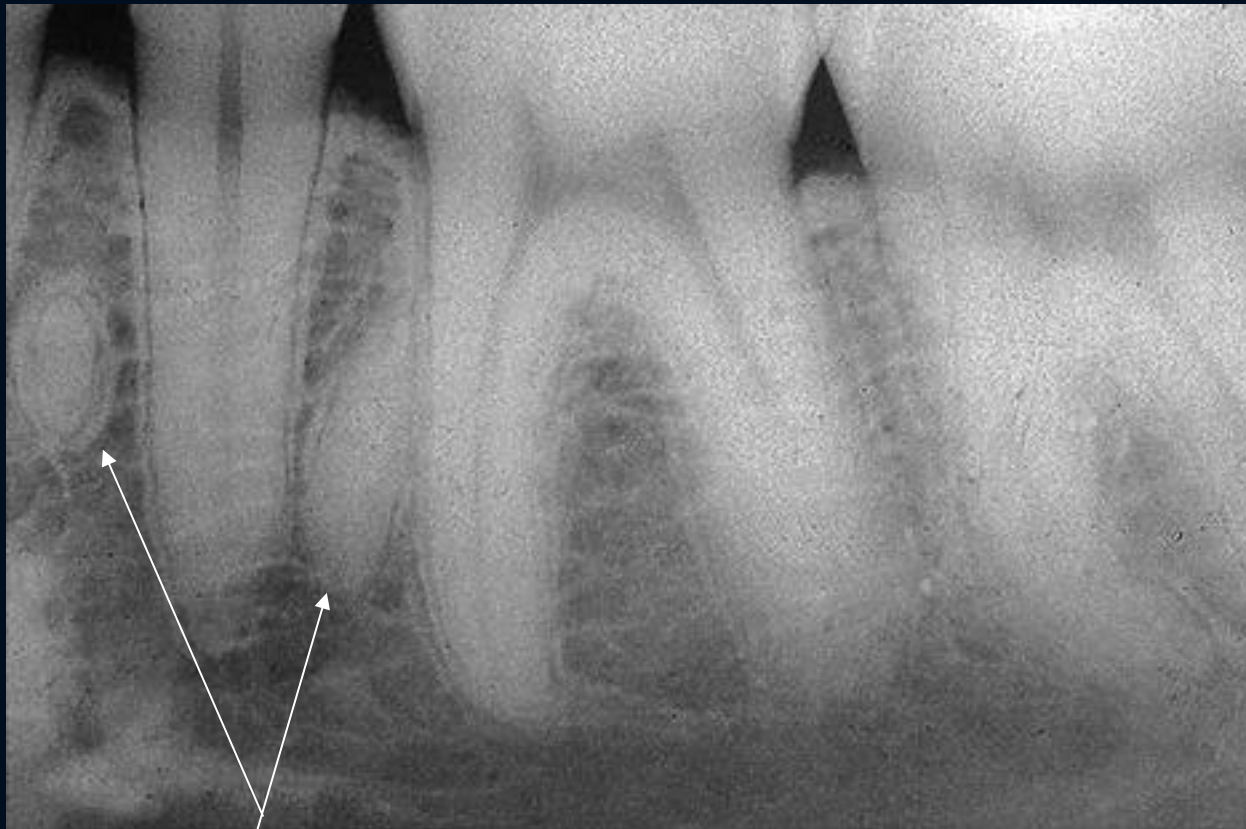
Parts of the roots of deciduous teeth which are not in the path of erupting permanent teeth may escape resorption.



e.g. The lower **E** have widely divergent roots, where the mesiodistal diameter of lower 5 is smaller than the distance between the roots of lower E, so part of roots remain deep in the bone

3. Remnants of deciduous teeth





Retained deciduous root tips



Incomplete physiologic root resorption resulting in retained root tips of deciduous second molar.

Their fate is:

- a. Surrounded by cellular cementum.**
- b. Ankylosed to bone.**
- c. Resorb and disappear.**
- d. Exfoliated.**

References:

1- Essentials of Oral Histology and Embryology, James K Avery; 3rd edition, Mosby. Chapter 6, Pages 81- 95

2- TenCate's Oral Histology. Development, structure and function, Antonio Nanci, 8th edition, 2008, Mosby Inc. Chapter 10, Pages 268-289

